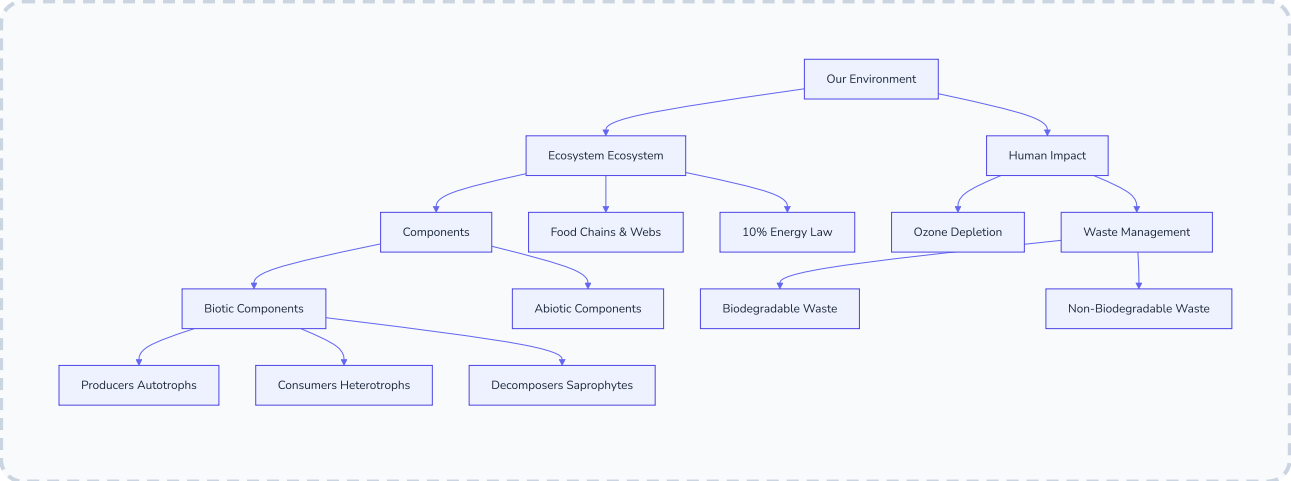
Chapter 13: Our Environment 🌍 🍃

Everything that surrounds us forms our environment. It includes both living (biotic) and non-living (abiotic) components. Understanding how energy flows through these components and the impact of human activities is crucial for our survival.

Master Mind Map 🗺️



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1. What is an Ecosystem? 🌱

An **Ecosystem** is a self-sustaining structural and functional unit of the biosphere consisting of a community of living beings and the physical environment, both interacting and exchanging materials between them.

- **Natural Ecosystems:** Forests, Ponds, Lakes, Oceans.
- **Artificial (Man-made) Ecosystems:** Gardens, Crop fields, Aquariums.

Components of an Ecosystem

1. **Abiotic Components (Non-living):** Physical factors like temperature, rainfall, wind, soil, and minerals.
2. **Biotic Components (Living):**
   - **Producers:** Organisms that can make their own food using sunlight (Photosynthesis). *Examples: Green plants, blue-green algae.*
   - **Consumers:** Organisms that depend entirely on producers for their food. *Divided into: Herbivores (plant-eaters), Carnivores (meat-eaters), Omnivores (both), and Parasites.*
   - **Decomposers:** Microorganisms that break down dead and decaying organic matter into simple inorganic substances, returning them to the soil. *Examples: Bacteria, Fungi.*

> **IMPORTANT POINTS** 📌
>
> Without **Decomposers**, the earth would become a giant garbage dump of dead plants and animals. They act as the natural cleaners and recyclers of the environment!

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2. Food Chains and Food Webs 🔗

A **Food Chain** is a linear sequence of organisms through which nutrients and energy pass as one organism eats another. *Example: Grass $\rightarrow$ Deer $\rightarrow$ Lion*

A **Food Web** is a complex network of many interconnected food chains within an ecosystem. It shows that most organisms eat, and are eaten by, more than one species. This provides stability to the ecosystem.

Trophic Levels

Each step or level of the food chain is called a **Trophic Level**.

1. **First Trophic Level:** Producers (Plants).
2. **Second Trophic Level:** Primary Consumers (Herbivores).

3. **Third Trophic Level:** Secondary Consumers (Small Carnivores).
4. **Fourth Trophic Level:** Tertiary Consumers (Top Carnivores like Tigers/Eagles).

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3. Flow of Energy and the 10% Law ⚡

The flow of energy in an ecosystem is strictly **Unidirectional** (it only flows in one direction: from the sun $\rightarrow$ producers $\rightarrow$ consumers). Energy given out as heat to the environment is never returning!

The 10% Law: Discovered by Raymond Lindeman, this law states that only **10%** of the energy entering a particular trophic level is available for transfer to the next higher trophic level.

- *Why?* Because 90% of the energy is lost as heat to the environment or used up by the organism for life processes (respiration, digestion, growth).
- **Result:** Food chains generally only consist of 3 or 4 trophic levels because the energy available for a 5th level would be too negligible to sustain life!

> ⓘ **NOTE** 📝
>
> Green plants capture only about **1%** of the total sunlight energy that falls on their leaves to convert into food!

Biological Magnification ☠️

The phenomenon where the concentration of harmful, non-biodegradable chemicals (like DDT pesticides) increases at each successive trophic level. Since humans are at the top of any food chain, we accumulate the maximum concentration of these harmful chemicals in our bodies!

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4. How do our Activities Affect the Environment? 🏭

A. Depletion of the Ozone Layer 🛡️

- **What is Ozone (O_3)?** A deadly poisonous gas at ground level, but in the stratosphere, it forms a protective shield that absorbs harmful Ultraviolet (UV) radiation from the Sun.
- **UV Radiation effects:** Causes skin cancer, cataracts, and damages crops.
- **Ozone Depletion:** Caused largely by synthetic chemicals like **Chlorofluorocarbons (CFCs)** which were used in refrigerators and fire extinguishers.
- In 1987, the United Nations Environment Programme (UNEP) successfully forged an agreement to freeze CFC production globally (The Montreal Protocol).

B. Managing the Garbage We Produce 🗑️

We generate massive amounts of waste daily.

1. **Biodegradable Waste:** Substances that can be broken down by biological processes (action of bacteria/fungi). *Examples: Food waste, paper, cotton.*
2. **Non-Biodegradable Waste:** Substances that cannot be broken down by biological processes. They persist in the environment for a long time and cause severe pollution. *Examples: Plastics, glass, synthetic fibers, electronic waste.*

Solutions: Better waste segregation, recycling plastics/metals, composting wet waste, and avoiding single-use plastics.

Quick Revision Notes 📌

1. **Ecosystem:** Combining the biological community with the physical environment.
2. **Decomposers:** Nature's vital recycling agents.
3. **Unidirectional Energy:** Energy from the Sun flows into plants and up the food chain, but never backwards.
4. **10% Law:** Only 10% energy transfers to the next level. Limits food chains to 3-4 steps.
5. **Biological Magnification:** Poisons concentrate more as they go UP the food chain. Top predators hold the highest poison loads.

6. **Ozone Layer:** O_3 blocks UV. CFCs destroy it.

Chapter Summary 🏁

Our environment features a delicate balance between biotic organisms and abiotic factors functioning as Ecosystems. Energy drives everything, but due to the 10% Law, energy drastically depletes as it moves up Food Chains. While energy dissipates, dangerous man-made toxins do the opposite: they concentrate and poison top predators through Biological Magnification. Furthermore, human inventions like CFCs have threatened our planet's protective sunscreen (the Ozone Layer), and non-biodegradable plastics are choking our Earth. Sustainable living, recycling, and ending CFC usage are vital to protecting the only home we have!